

Ian Leung

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EDUCATION

University of Waterloo

Sep 2025 – Present

Bachelor of Computer Science and Finance - Double Major

- President's Scholarship of Distinction (\$5000), Undergraduate Achievement Award (\$1500)
- Statistics Minor

Relevant Coursework: Functional Programming (Advanced), Interpreters (Advanced), Linear Algebra, Calculus, Probability, Proofs, Financial Markets and Data Analytics

TECHNICAL SKILLS

Languages Python, Java, C++, C#, C, Go, Kotlin, TypeScript, SQL, HTML, CSS, Bash
Technologies Temporal, Pytorch, Supabase, Firebase, MongoDB, React, Maven, Qt, Openrouter, OpenCV
Developer Tools Linux, Git, Docker, Jupyter, DigitalOcean, Modal, Render, Vercel, Unity, Android Studio

WORK EXPERIENCE

Roblox

San Mateo, CA

Software Engineer Intern

Jun 2026 – Present

- Developing an automated CI/CD pipeline for network configuration deployments using **GitHub Actions** and **Temporal**, replacing manual rack-switch config push workflows with a safer, auditable rollout system
- Designing a canary-first rollout strategy with pre/post-deployment health checks, progressive rollout gates, and staged batch expansion to safely validate and deliver network changes to **100+ million** daily active users
- Implementing deployment safety mechanisms including device locking, validation gates, and rollback logic to prevent unsafe configuration pushes during canary and batch rollouts

Sunnybrook Research Institute

Toronto, ON

Software Research Intern

Jul 2025 – Aug 2025

- Built an event tracing and simulation tool to simulate medical professionals' interactions with **Focused Ultrasound**  software, enabling realistic testing and diagnosis of software defects, unintended user behaviours, and edge cases
- Created an automated software testing framework with **PyQt** and **Pydantic**, generating synthetic and replayable JSON event flows, reducing software testing times by **91%**
- Implemented a logging API using **Diffib** and **Loguru** to validate test results against baselines, identifying **5+** bugs

Data Science Club

Waterloo, ON

Software Developer & Hackathon Organizer

Sep 2025 – May 2026

- Maintaining **www.uwdatascience.ca** - UWDSOC's primary website and associated apps for events and internal tooling
- Created an internal onboarding form and API to manage executive profiles using **TypeScript**, **React** and **Supabase**
- Managed food and logistics for 350+ hackathon attendees at **CxC**  - Canada's largest student-run AI hackathon

PROJECTS

Pharmagotchi: Pharmacy Companion App | Android Studio, Figma, Kotlin, XML, OpenRouter

- Built an **Android app** featuring a virtual pharmacy companion that provides users with health tracking, medication reminders, AI-powered advice, and direct communication channels with healthcare providers
- Leveraged **OpenRouter LLMs** to integrate a virtual assistant for personalized health advice, symptom checking, and medication information, allowing users to better understand their health and medications
- Generates automated health reports and medication summaries for users to easily share with healthcare providers

PeerAssist - Full-Stack Peer Reviewing | Java, MongoDB, Apache PDFBox, Maven

- Created a full-stack **Java** desktop application for students to upload, review and grade peers' academic work
- Integrated a **Supabase** backend for authentication and data storage, enforcing **Row-Level Security** policies across profiles, documents, and reviews tables to control per-user access
- Rendered uploaded PDFs with **Apache PDFBox** and added search, sorting, and ranking by average review score; packaged the app into a single zero-config executable JAR via **Maven** shade plugin

Trust No Ghost - 3D Maze Exploration Game | Unity, C#, Itch.io

- Shipped Trust No Ghost  for **60+ users** at the UW Game Jam Fall 2025, placing in the **top 10** most rated games
- Utilized Unity and C# to develop expanding level designs, AI pathfinding algorithms, and dynamic moving mechanics